

The Power of Summary Statistics

Before diving into complex analyses, smart traders first get a broad overview of their data. Summary statistics provide a snapshot of key characteristics—like examining a patient's vital signs before a detailed diagnosis.

Let's say we've collected Apple stock data and want to understand its basic properties:

```
1. import pandas as pd
2. import numpy as np
3. import yfinance as yf
4.
5. # Get Apple stock data
6. apple = yf.download('AAPL', start='2022-01-01', end='2023-01-01')
7.
8. # Calculate daily returns
9. apple['Returns'] = apple['Adj Close'].pct_change()
10.
11. # Get summary statistics
12. summary = apple['Returns'].describe()
13. print(summary)
14.
```

This simple code gives us critical insights:

```
count    251.000000
mean      0.000311
std       0.026014
min      -0.122370
25%      -0.013417
50%       0.001924
75%       0.016345
max       0.079376
```

These numbers tell a powerful story:

- The **mean** daily return is positive but very small (0.03%)
- The **standard deviation** (a measure of volatility) is about 2.6%